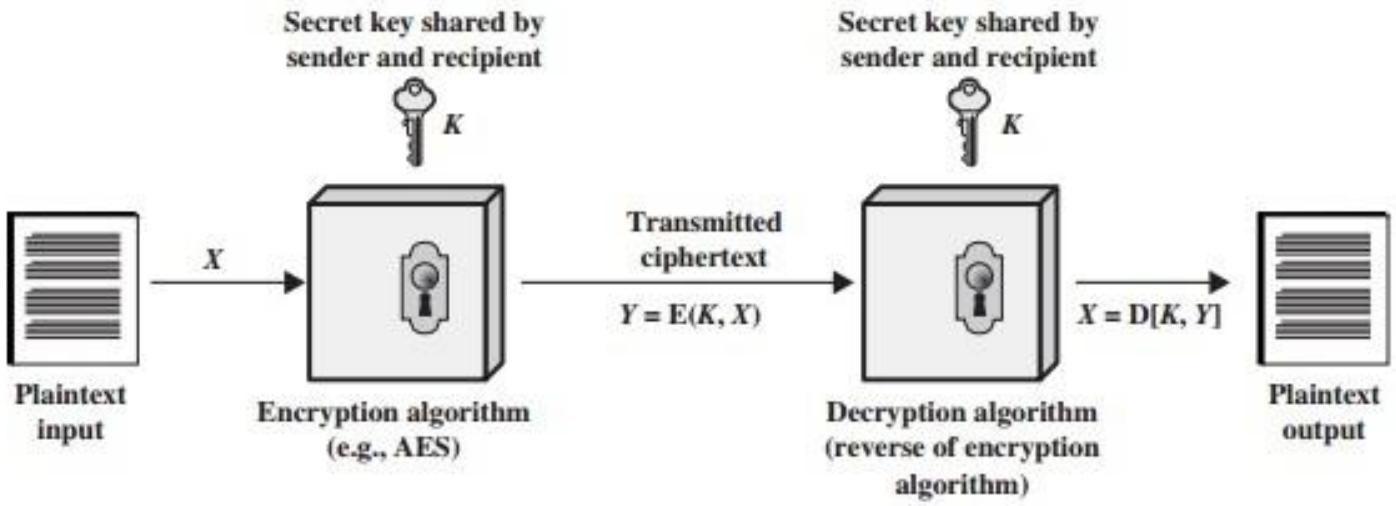
Symmetric Key Cryptography

Same key is used for both encryption and decryption process.

Symmetric cipher model consists of five elements:

- Plain Text
- Encryption Algorithm
- Secret Key
- Cipher Text
- Decryption Algorithm



Symmetric Key Cryptography

Plain Text: This is the original intelligible message or data that is fed into the algorithm as input.

Encryption algorithm: The encryption algorithm performs various substitutions and transformations on the plaintext.

Secret key: It is a value independent of the plain text and of the algorithm. The exact substitutions and transformations performed by the algorithm depend on the key.

Cipher text: It is the unreadable message produced as output. It depends on the plain text and secret key.

Decryption algorithm: This is essentially the encryption algorithm run in reverse. It takes the cipher text and the secret key and produces the original plaintext.

Symmetric Key Cryptography

Mathematical Representations:

Mathematically it is represented as Y=E(K,X),

Where Y=the cipher text,

E =encryption algorithm,

K =secret key and

X=plain text.

Same as X=D(K,Y)

Where X=plain text

D=decryption algorithm

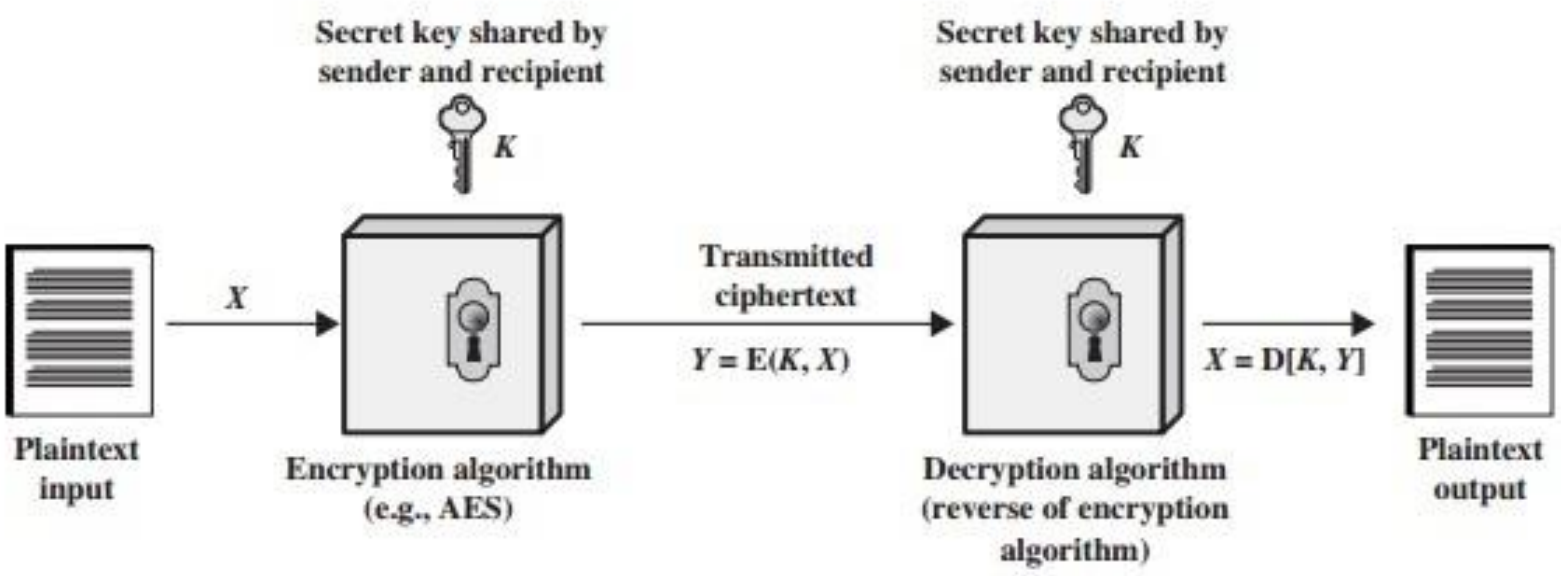
K=secret key

Y=cipher text.

The single key K is used for both encryption and decryption.

Symmetric Key Cryptography

- Here the possibility is that if an attacker get the cipher text, he/she may apply different permutations and combinations to decrypt and obtain the original plain text.
- So, always the sender has to think on applying different encoding technique on plain text and convert it into cipher text, therefore attacker cannot read the original message.



Symmetric Key Cryptography

Advantages of Symmetric key Cryptography:

Symmetric key is faster than asymmetric key cryptography.

Same key is used for both encryption and decryption. So, the receiver cannot decrypt the encrypted date without getting the key from the sender, or without the sender permission.

Symmetric key achieves the authentication principle, because it checks receivers identity.

System resources are less used in this type of cryptography.

Disadvantages:

Once key is stolen while transmitting data, encrypted data can be easily decrypted.

In symmetric key cryptography the key is transmitted first and then message is transferred to the receiver. If any attacker can intercept the communication he/she may decrypt the message easily.

Asymmetric Key Cryptography

Asymmetric key cryptography is also called public key cryptography.

Asymmetric cipher model consists of six elements:

- Plain text
- Encryption algorithm
- Private key
- Public key
- Cipher text
- Decryption algorithm

Asymmetric Key Cryptography

Plain Text: This is the original intelligible message or data that is fed into the algorithm as input.

Encryption algorithm: The encryption algorithm performs various transformations on the plaintext.

Public and Private keys: This is a pair of keys that have been selected so that if one is used for encryption, the other is used for decryption.

Cipher Text: Is the unreadable output produced as output of the encryption. It depends on the plain text and key. Two different keys produce two different outputs.

Decryption algorithm: It takes the cipher text and the secret key and produces the original plaintext.

Asymmetric Key Cryptography

Public key is distributed to all the users, and the private is known only by the particular user only.

There are two scenarios of encryption model:

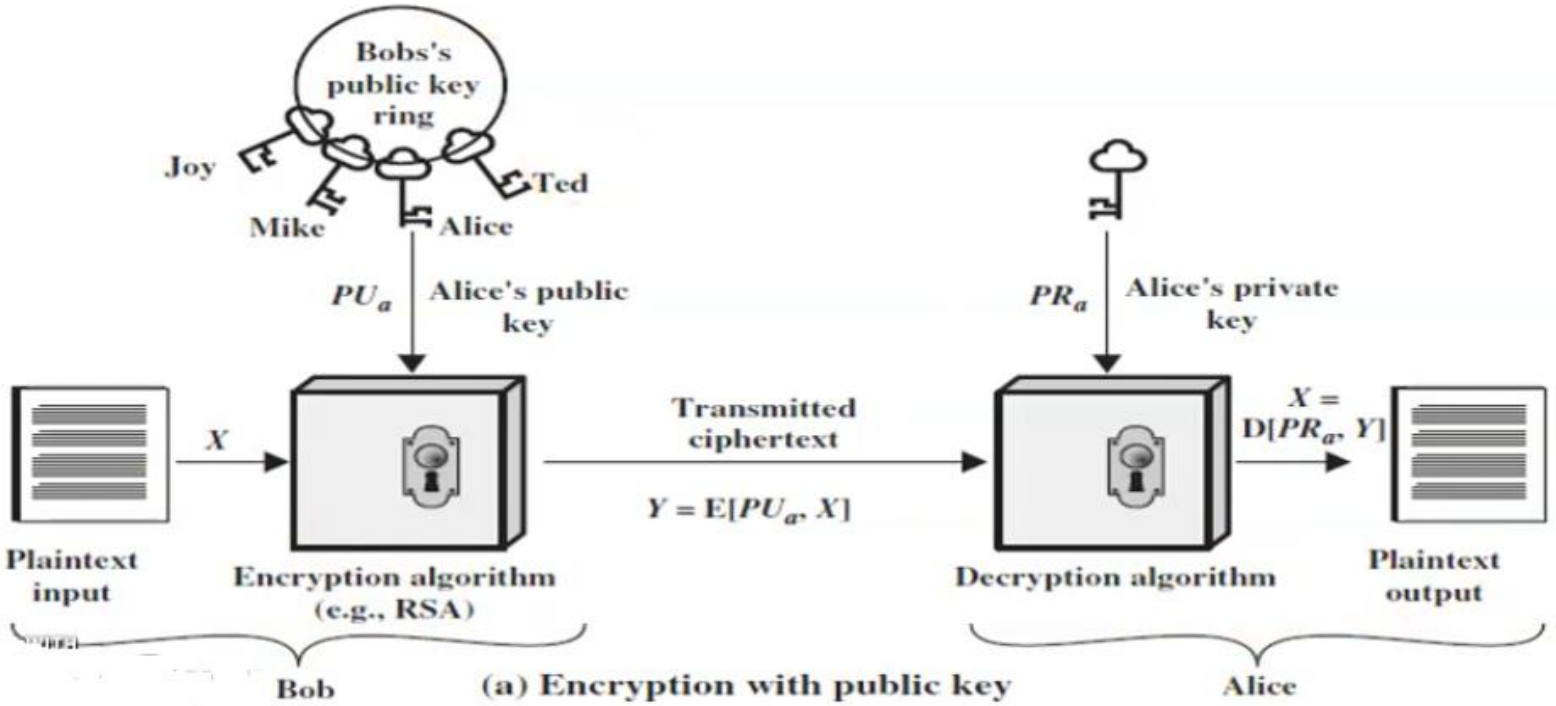
Public key is used for encryption and private key is used for decryption.

Private key is used for encryption and public is used for decryption.

Asymmetric Key Cryptography

Scenario-1: Encryption with public key:

- If Bob wants to send message to Alice, Bob must use the public key of Alice. Message to be transmitted after encryption of message using Alice's public key.
- After receiving the message, Alice can decrypt the message using only her private key.



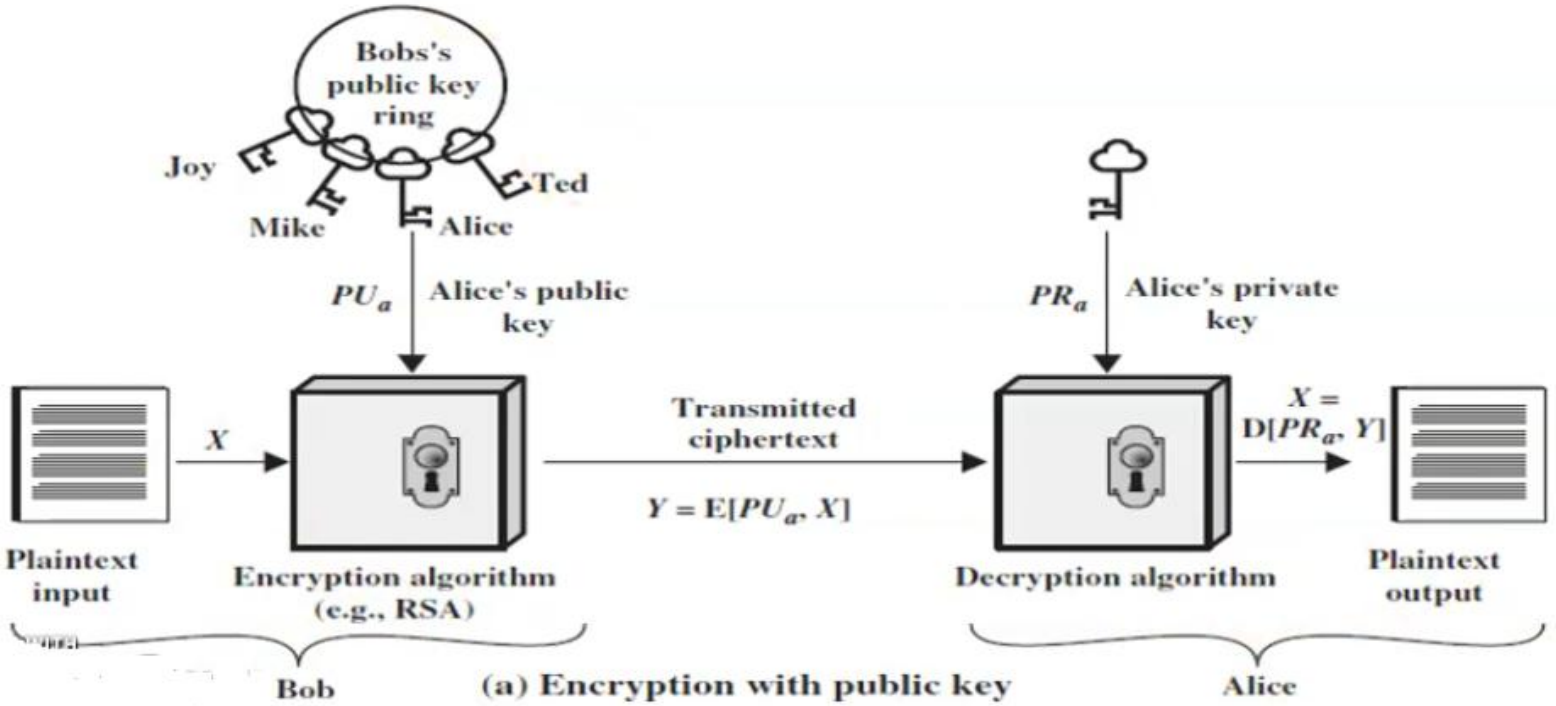
Asymmetric Key Cryptography

Scenario-1: Encryption with public key:

Y=E(Pub(A),X)

X=D(Pri(A),Y)

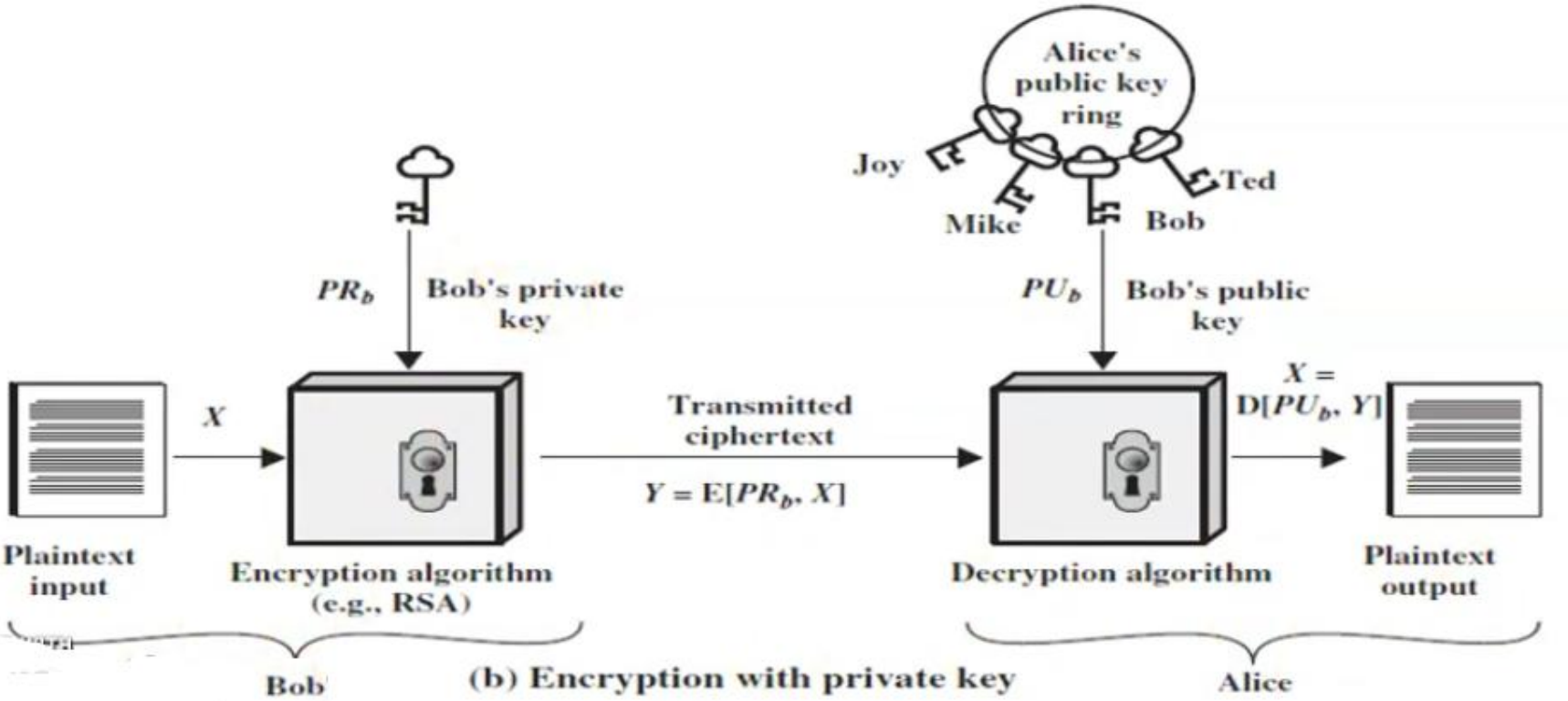
Where X=Plain Text, Y=Cipher Text, Pub(A)=Alice's public key, Pri(A)=Alice's rivate key.



Asymmetric Key Cryptography

Scenario-2: Encryption with Private key:

If Bob wants to send message to Alice, Bob must use his own private key to encrypt the message. After Transmission, Alice uses the public key of Bob to decrypt the ciphertext.



Asymmetric Key Cryptography

Scenario-2: Encryption with Private key:

Mathematically,

Y=E(Pri(B),X)

X=D(Pub(B),Y)

Where X=Plain text, Y=Cipher text, E=Encryption algorithm, D=decryption algorithm, Pri(B)=Bob's private key, and Pub(B)=Bob's public key

